Line integrals and double integrals. Polar coordinates.

[SUBMIT ANSWERS TO ALL QUESTIONS]

Line integral of the first kind: $\int\limits_C f(x,y,z)ds$; second kind: $\int\limits_C \boldsymbol{A}\cdot d\boldsymbol{r} = \int\limits_C (A_x dx + A_y dy + A_z dz)$.

Double integral: $\iint\limits_S f(x,y)dxdy$. Green's theorem: $\oint\limits_C (Pdx + Qdy) = \iint\limits_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy$.

Elements of length and area in polar coordinates $(r,\,\theta)$: $ds = \sqrt{dr^2 + r^2 d\theta^2}$, $dS = rdrd\theta$.

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1. Find the line integral of the first kind $\int\limits_C xyds$, where C is a quarter of the circle $x^2+y^2=a^2$ lying in the first quadrant. [Hint: use the parameterisation $x = a\cos t$, $y = a\sin t$.] [20]

2. Find $\int\limits_C \frac{ds}{x^2+y^2+z^2}$, where C is the turn of the helix $x = a\cos t$, $y = a\sin t$, $z = bt$ from $t=0$ to $t=2\pi$. [Hint: use substitution $u = bt/a$ to obtain a standard integral.] [20]

3. Evaluate the following double integrals:

$$\text{(a)} \int_0^2 dy \int_0^1 (x^2+2y)dx, \quad \text{(b)} \int_3^4 dx \int_1^2 \frac{dy}{(x+y)^2}, \quad \text{(c)} \int_0^{2\pi} d\theta \int_{a\sin\theta}^{a} rdr.$$

[20]

4. Write the equations of curves bounding the integration regions of the following double integrals, and sketch them.

$$\text{(a)} \int_0^4 dx \int_{x/4}^{1} f(x,y)dy, \quad \text{(b)} \int_0^3 dx \int_{x^2}^{3x} f(x,y)dy, \quad \text{(c)} \int_0^2 dx \int_0^{\sqrt{9-x^2}} f(x,y)dy,$$

[20]

5. Show that for $P=-y$, $Q=0$ the integral on the left-hand side of Green's theorem gives the area bounded by curve C. Hence, use this line integral to find the area of the ellipse

$$x = a\cos t, \quad y = b\sin t \quad (0 \le t \le 2\pi).$$

[20]

MTH1021 Problem Sheet 20 Solutions.

Line Integrals, Double Integrals, Polar Co-ords.

Q1.

$$x = a\cos t \qquad y = a\sin t$$
$$x'(t) = -a\sin t \qquad y'(t) = a\cos t.$$

$$ds = \sqrt{[x'(t)]^2 + [y'(t)]^2}\,dt$$
$$= \sqrt{(-a\sin t)^2 + (a\cos t)^2}\,dt$$
$$= a\,dt.$$

Hence

$$\int_C xy\,ds = \int_0^{\pi/2} a\cos t\, a\sin t\, a\,dt$$
$$= a^3\int_0^{\pi/2} \sin t\cos t\,dt$$
$$= a^3\int_0^{\pi/2} \sin t\, d(\sin t)$$
$$= \frac{a^3}{2}\left[\sin^2 t\right]_0^{\pi/2}$$
$$= \frac{a^3}{2}\left[1 - 0\right]$$
$$= a^3/2.$$

* The limits $t=0$ and $t=\pi/2$ correspond to the starting point $(a,0)$ and the end point $(0,a)$ of the arc required *

Q2.

$$x = a\cos t \qquad y = a\sin t \qquad z = bt$$
$$x'(t) = -a\sin t \qquad y'(t) = a\cos t \qquad z'(t) = b.$$

$$ds = \sqrt{[x'(t)]^2 + [y'(t)]^2 + [z'(t)]^2}\,dt$$
$$= \sqrt{(-a\sin t)^2 + (a\cos t)^2 + b^2}\,dt$$
$$= \sqrt{a^2+b^2}\,dt$$
= the element of length.

Hence

$$\int_C \frac{ds}{x^2+y^2+z^2} = \int_0^{2\pi} \frac{\sqrt{a^2+b^2}\,dt}{a^2\cos^2 t + a^2\sin^2 t + b^2 t^2}$$

$$= \sqrt{a^2+b^2} \int_0^{2\pi} \frac{dt}{a^2+b^2t^2}$$

$$= \sqrt{a^2+b^2} \int_0^{2\pi} \frac{1}{a^2\left(1+\frac{b^2}{a^2}t^2\right)} dt.$$

$$= \frac{\sqrt{a^2+b^2}}{a^2} \int_0^{2\pi} \frac{dt}{1+\left(\frac{bt}{a}\right)^2}.$$

let $u = \frac{bt}{a} \Rightarrow du = \frac{b}{a}\,dt \Rightarrow \frac{a}{b}\,du = dt.$

$$\therefore \int \frac{dt}{1+\left(\frac{bt}{a}\right)^2} \longrightarrow \frac{a}{b}\int \frac{du}{1+u^2}$$

We can now use the standard integral

$$\int \frac{du}{1+u^2} = \arctan u + C$$

$$\therefore \int \frac{dt}{1+\left(\frac{bt}{a}\right)^2} = \arctan\left(\frac{bt}{a}\right) + C.$$

Hence we have

$$\frac{\sqrt{a^2+b^2}}{a^2} \cdot \frac{a}{b} \left[\arctan\left(\frac{bt}{a}\right) \right]_0^{2\pi}$$

$(\arctan 0 = 0)$.

$$= \frac{\sqrt{a^2+b^2}}{ab} \arctan \frac{2\pi b}{a}$$

(3).

(a) $\int_0^2 dy \int_0^1 (x^2+2y)\,dx.$

$= \int_0^2 dy \left[\frac{x^3}{3} + 2xy\right]_0^1 = \int_0^2 \left(\frac{1}{3} + 2y\right) dy$

$= \left[y/3 + y^2\right]_0^2 = \frac{2}{3} + 4 - 0 = 4\,^2\!/_3.$

(b). $\int_3^4 dx \int_1^2 \frac{dy}{(x+y)^2}$

$= \int_3^4 dx \left[-\frac{1}{(x+y)}\right]_1^2$

remember x is treated as a constant here!

$= \int_3^4 \left(\frac{1}{x+1} \;\; \frac{-1}{x+2}\right) dx.$

$= \left[\ln(x+1) - \ln(x+2)\right]_3^4$

$= \ln 5 - \ln 6 - \ln 4 + \ln 5$

$= \ln \frac{5.5}{6.4} = \ln \frac{25}{24}.$

(c). $\int_0^{2\pi} d\theta \int_{a\sin\theta}^{a} r\,dr$

$= \int_0^{2\pi} d\theta \left[\frac{r^2}{2}\right]_{a\sin\theta}^{a} = \int_0^{2\pi} d\theta \left[\frac{a^2}{2} - \frac{a^2\sin^2\theta}{2}\right]$

$= \frac{1}{2}a^2 \int_0^{2\pi} (1-\sin^2\theta)\,d\theta = \frac{1}{2}a^2 \int_0^{2\pi} \cos^2\theta\,d\theta$

Using the double-angle formula $\cos 2\theta = 2\cos^2\theta - 1 \Rightarrow \cos^2\theta = \frac{1}{2}(1+\cos 2\theta)$

$= \frac{1}{4}a^2\left[\theta + \frac{1}{2}\sin 2\theta\right]_0^{2\pi}$

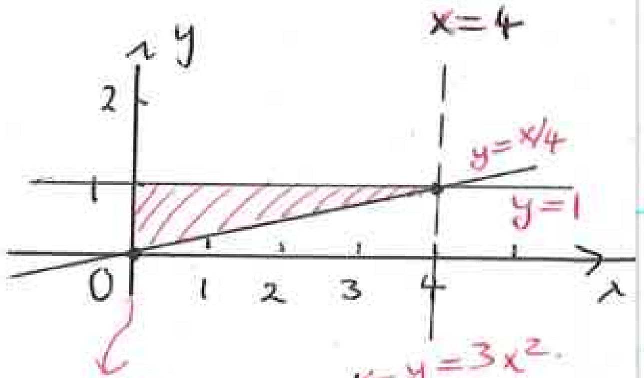
$= \frac{1}{4}a^2\left[2\pi + \frac{1}{2}\underbrace{\sin 4\pi}_{=0} - 0\right] = \frac{\pi a^2}{2}.$

Q4.

(a) $\int_0^4 dx \int_{x/4}^{1} f(x,y)\,dy$

$0 \le x \le 4$, $x/4 \le y \le 1$

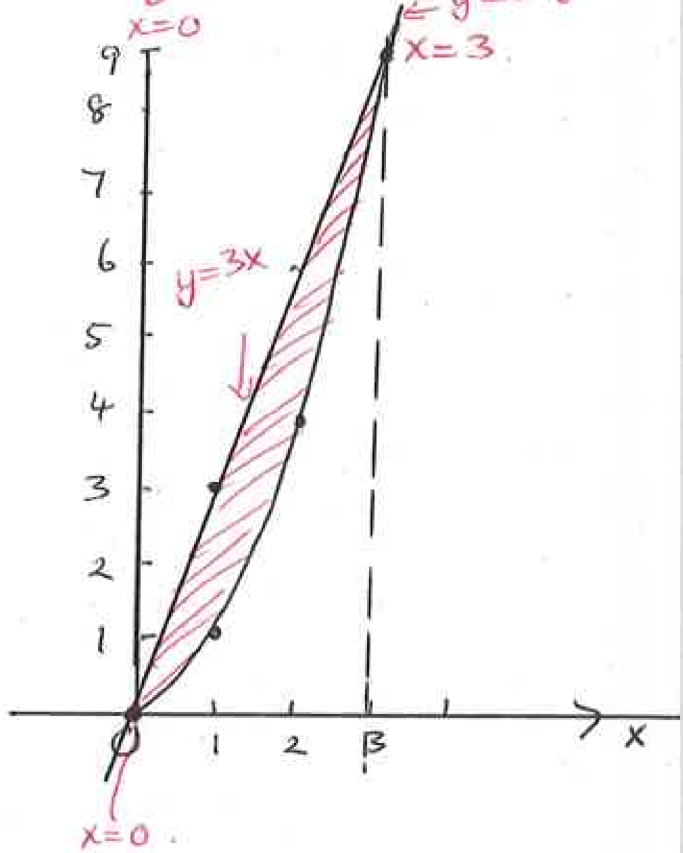
The integration region is bounded by the lines $x=0$, $x=4$, $y=x/4$ and $y=1$.



(b) $\int_0^3 dx \int_{x^2}^{3x} f(x,y)\,dy$

$0 \le x \le 3$, $x^2 \le y \le 3x$

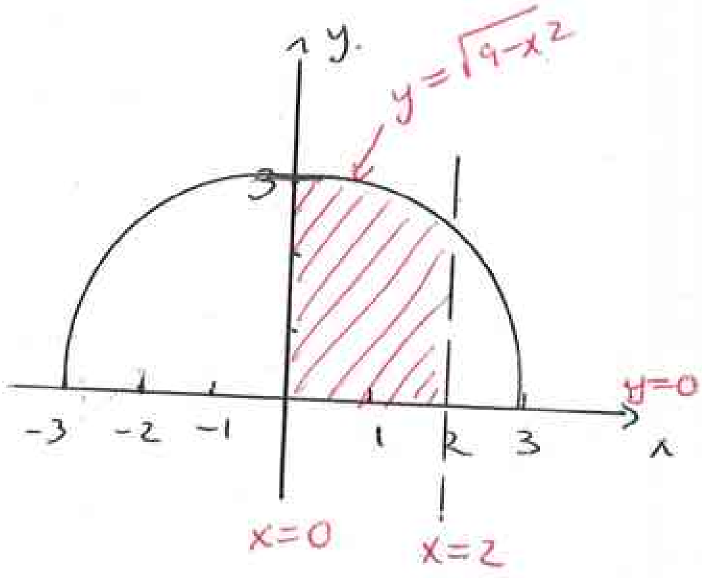
The integration region is bounded by the lines $x=0$, $x=3$, a parabola $y=x^2$ and the line $y=3x$.



(c) $\int_0^2 dx \int_0^{\sqrt{9-x^2}} f(x,y)\,dy$

$0 \le x \le 2$, $0 \le y \le \sqrt{9-x^2}$

The integration region is bounded by $x=0$, $x=2$, $y=0$ and $y=\sqrt{9-x^2}$. the latter is an arc of the circle $x^2+y^2=9$ of radius 3, with centre at the origin.



Q5. Green's theorem states that

$$\int_C P\,dx + Q\,dy = \iint_S \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

$P = -y \qquad Q = 0$

$\Rightarrow \dfrac{\partial P}{\partial y} = -1 \qquad \Rightarrow \dfrac{\partial Q}{\partial x} = 0$

and the right hand side becomes

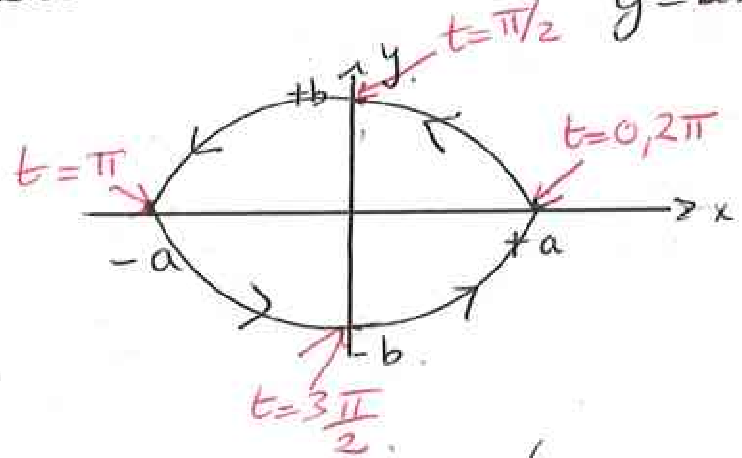
$$\iint_S (0 - (-1))\,dx\,dy = \iint_S dx\,dy$$

which is the area of the integration region S. Hence this can be found from the linear integral on the left-hand side.

$$\int_C P\,dx + Q\,dy = -\int_C y\,dx. = A \text{ say}$$

where C is the curve that bounds S and is traversed in the anticlockwise direction.

Lets sketch the ellipse $x = a\cos t$, $y = b\sin t$ $\quad 0 \le t \le 2\pi$



Values of a and b (for $a > b$) are known as the semimajor and semiminor axes of the ellipse.

The area $A = -\int_C y\,dx.$

$y = b\sin t$

$x = a\cos t \Rightarrow dx = -a\sin t\, dt.$

$$\therefore A = -\int_0^{2\pi} b\sin t\,(-a\sin t)\, dt.$$

$$= ab\int_0^{2\pi} \sin^2 t\, dt$$

Using the double angle formula

$\cos 2t = 1 - 2\sin^2 t \Rightarrow \sin^2 t = \frac{1}{2}(1-\cos 2t)$

$$\therefore A = \frac{ab}{2}\int_0^{2\pi}(1-\cos 2t)\, dt$$

$$= \frac{ab}{2}\left[t - \frac{1}{2}\sin 2t\right]_0^{2\pi}$$

$$= \frac{ab}{2}\left[2\pi - \frac{1}{2}\underbrace{\sin 4\pi}_{=0} - 0\right]$$

$$= \pi ab$$

= area of the given ellipse

<u>check:</u> For $b=a$ we have a circle of radius a and the area becomes $A=\pi a^2$ which is correct.